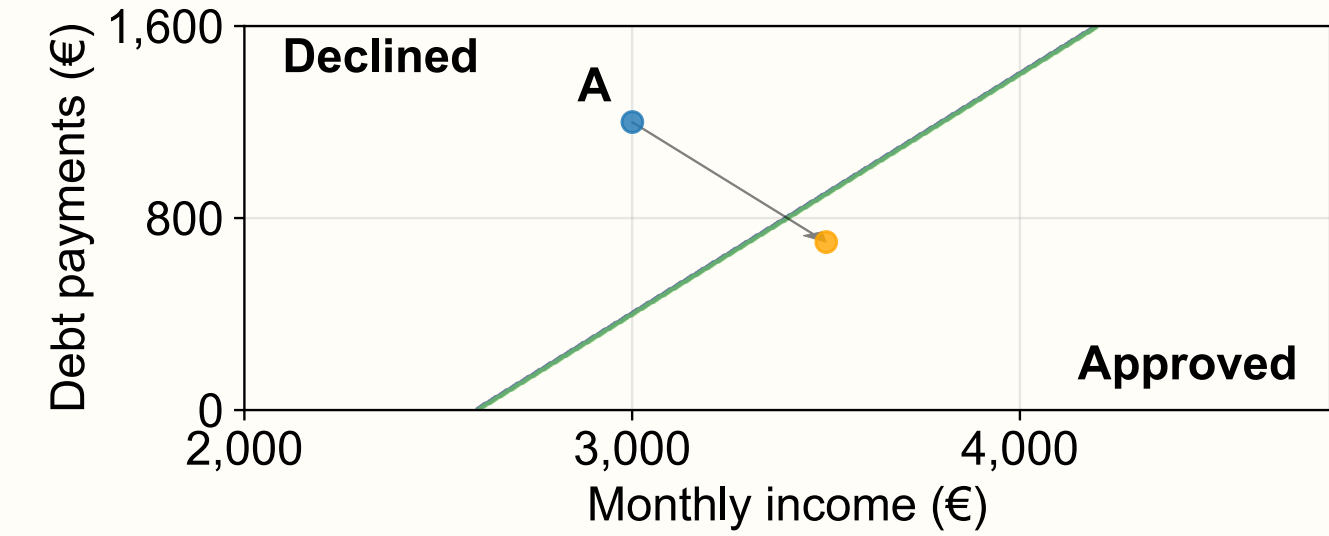


What is a counterfactual explanation?
A small input change that flips a model's prediction.

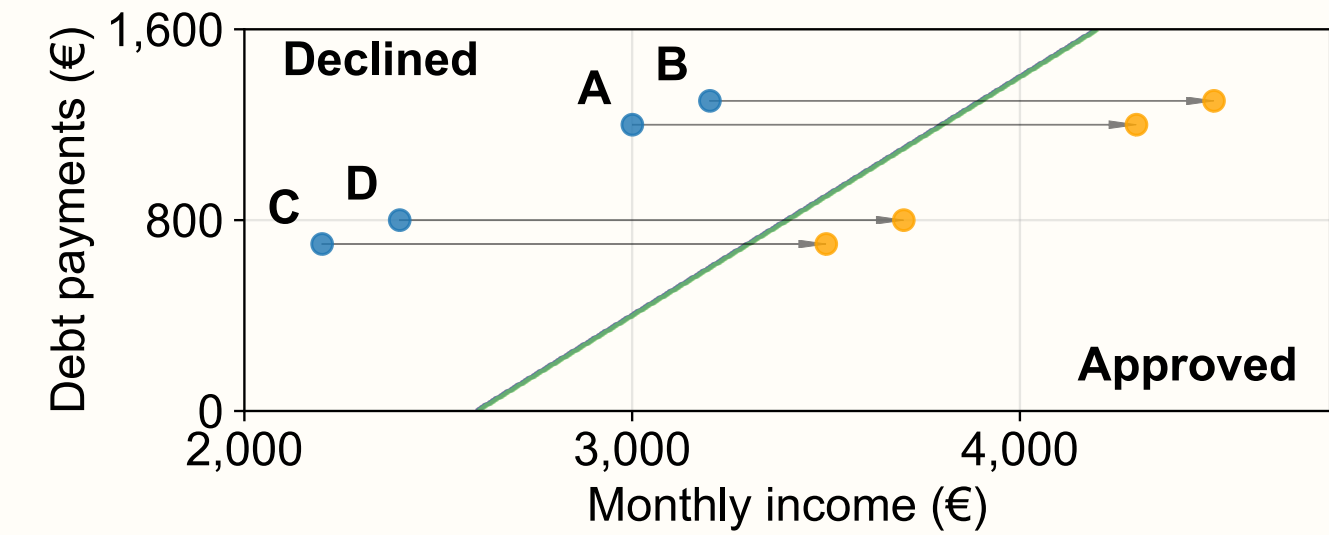
LOAN APPLICATION EXAMPLE

Local	One applicant	
€/month	Original A	Counterfactual
Monthly income	€3,000	€3,500
Debt payments	€1,200	€700
Employment	Full-time	Full-time
Prediction	Declined	Approved



Global One shared change

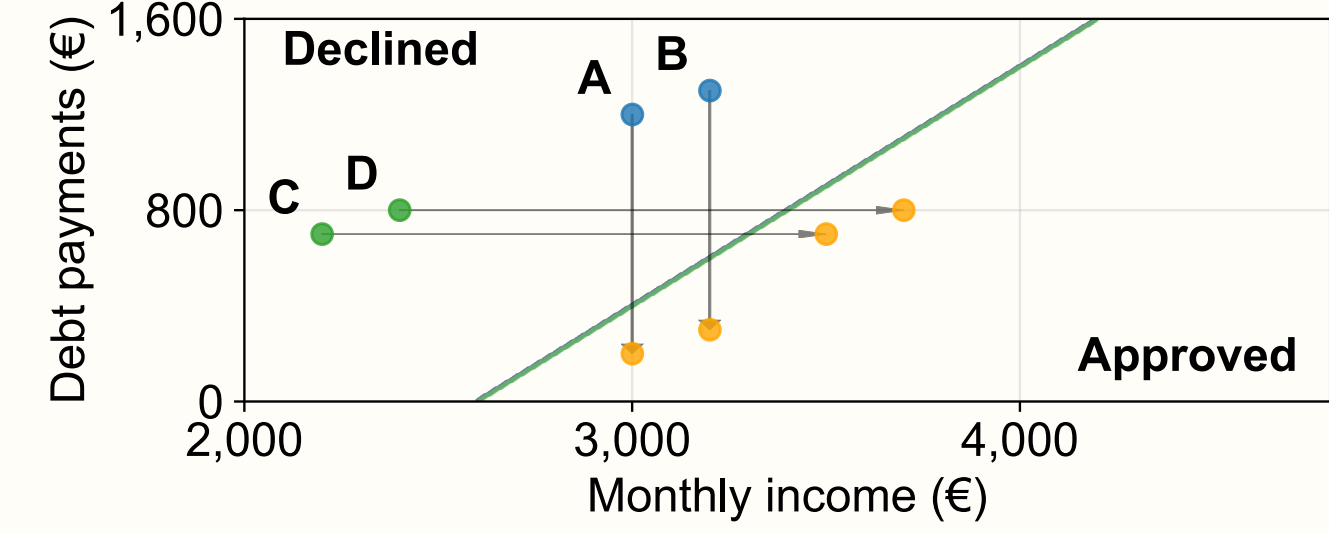
A-D Income +€1,300; debt unchanged.



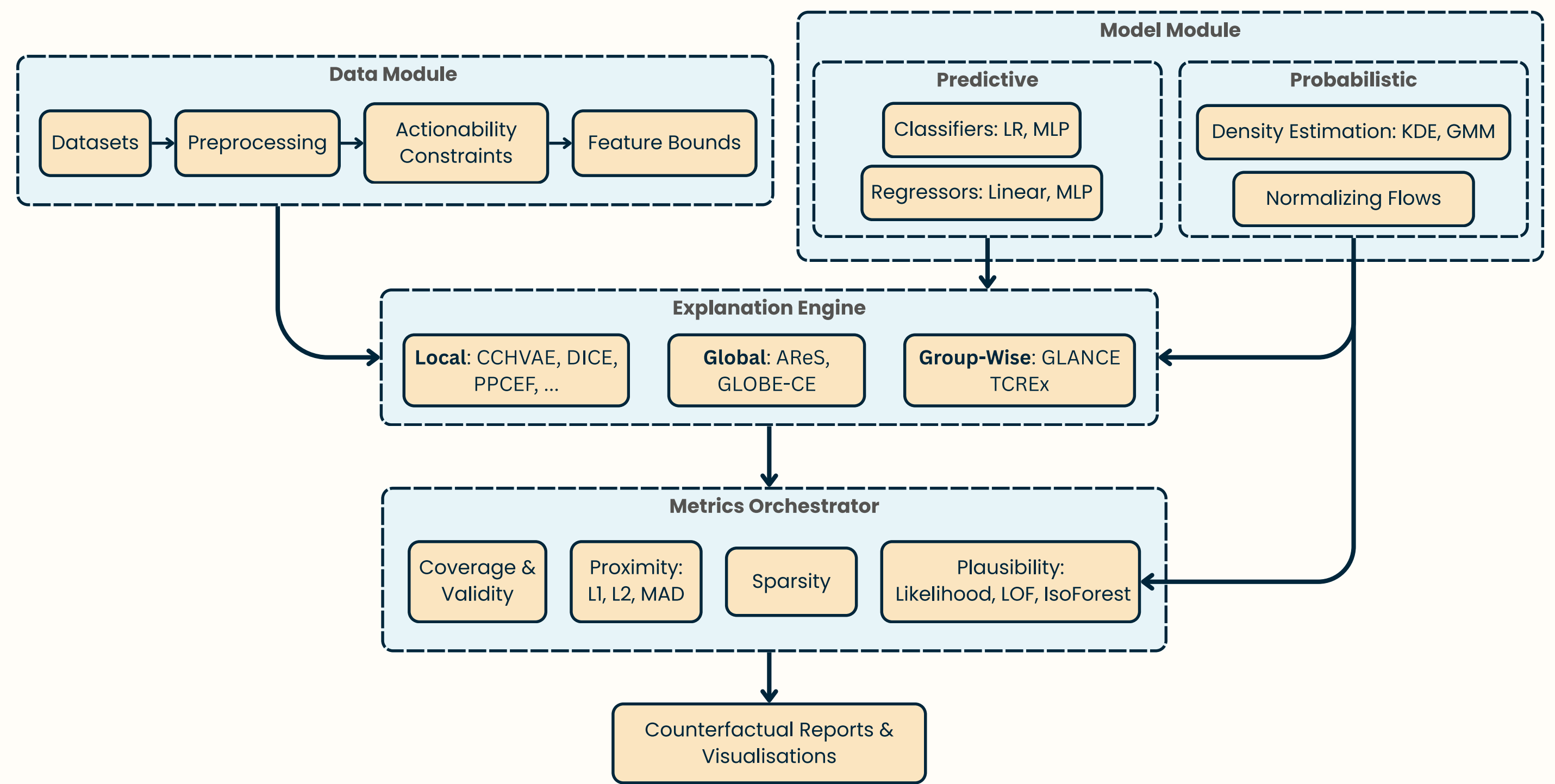
Group-wise One change per group

A-B Debt -€1,000; income unchanged.

C-D Income +€1,300; debt unchanged.



● Original ● Counterfactual — Decision boundary



18 Datasets

Classification
Adult Census · Audit · Bank Marketing · Blobs · Credit Default · Digits · German Credit · Give Me Some Credit · HELOC · Law · Lending Club · Moons · Wine
Regression
Synthetic · Concrete · Diabetes · Yacht · SCM20D

14 Methods

Local
Wachter · Artelt · DiCE · CCHVAE · PPCEF · CEM · CEGP · CADEX · SACE · CEARM
Global
ARes · GLOBE-CE
Group-wise
GLANCE · T-CREx

2 Backbones / Task

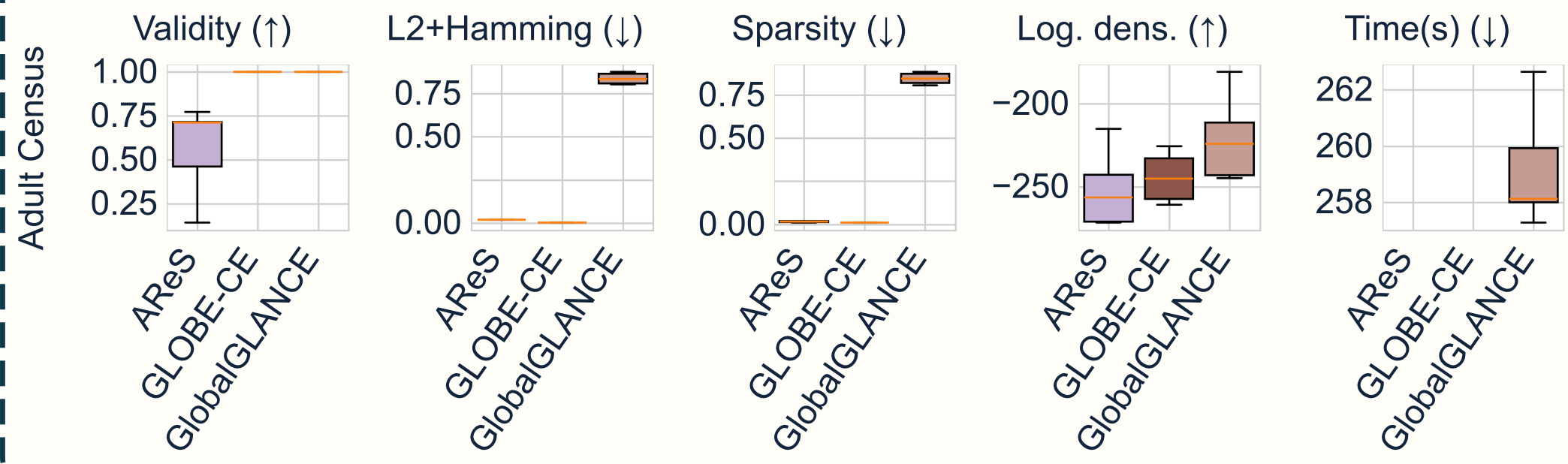
Classification
Multi-Layer Perceptrons (MLP) · Logistic Regression
Continuous targets
Linear Regression · MLP Regressor

9 Classification Metrics

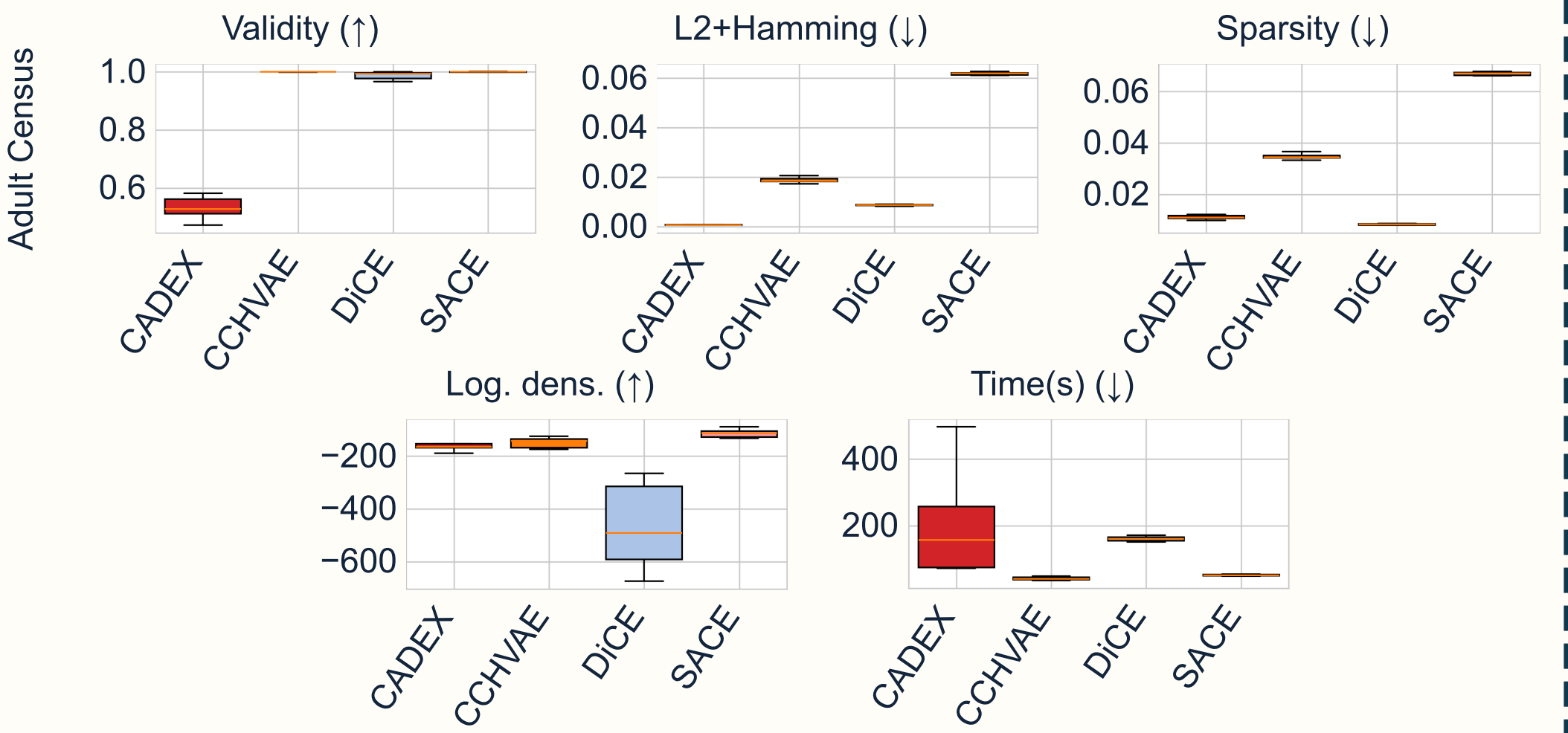
Coverage · Validity · Sparsity · Prob. plausibility · Log-density · LOF · Isolation Forest · Proximity (L2 / L2-Hamming) · Runtime

Results

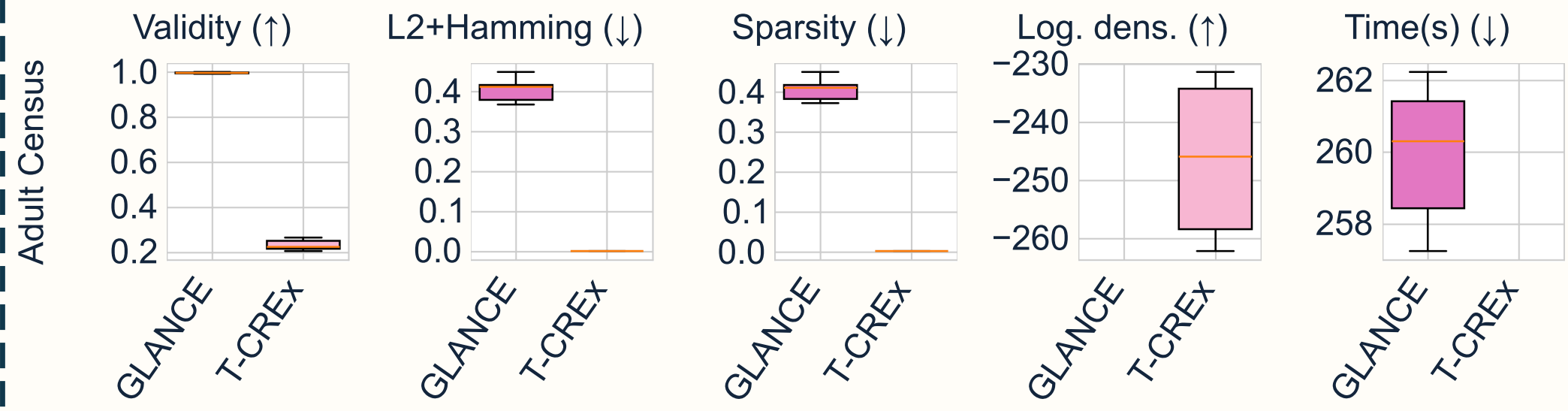
Global: validity and failures



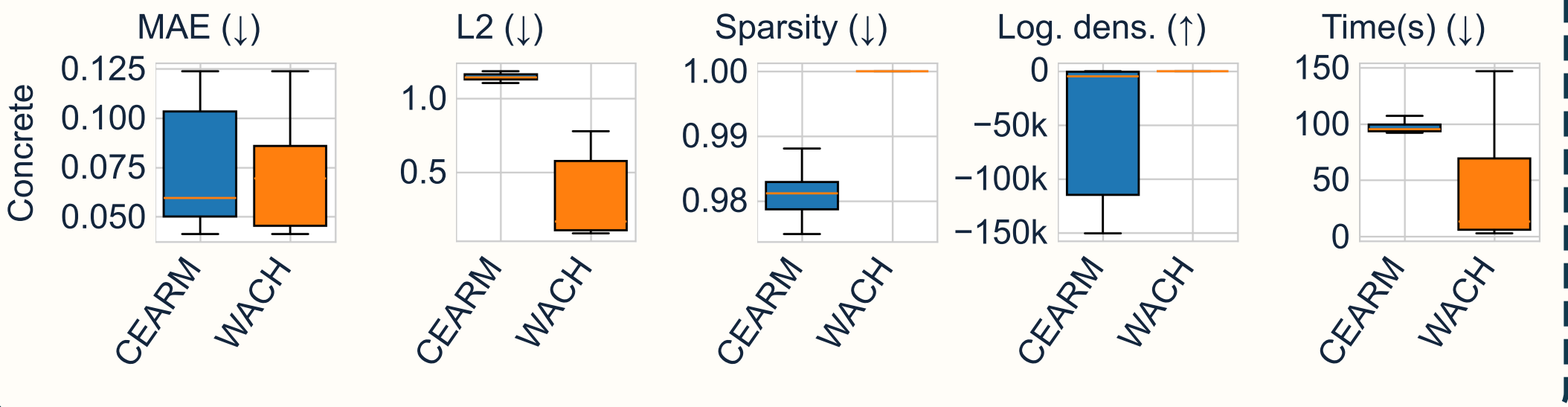
Local: no single winner



Group-wise: reach versus change



Regression: accuracy versus change



Three contributions

CONTROL
Controlled evaluation protocol

COVER
Broad benchmark across CE paradigms

EXTEND
Extensible open-source benchmark workbench

